

# Multinomial Discrete Choice: Mixed Logit

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Chris Conlon

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## Mixed/ Random Coefficients Logit

An alternative is to allow for individuals to have **random coefficients**  $\beta_i$ , mixing over various logits:

$$\begin{aligned}\beta_i &= \bar{\beta} + \Pi y_i + \Sigma \nu_i & \mu_{ij} &= x_j \cdot (\Pi y_i + \Sigma \nu_i) \\ u_{ijt} &= x_j \beta_i + \varepsilon_{ij} & &= x_j \beta + \mu_{ij} + \varepsilon_{ij} \\ \sigma_j(\theta) &= \int \frac{\exp[x_j \beta_i]}{1 + \sum_k \exp[x_k \beta_i]} f(\beta_i | \theta) & &= \int \frac{\exp[x_j \beta + \mu_{ij}]}{1 + \sum_k \exp[x_k \beta + \mu_{ik}]} f(\mu_i | \theta)\end{aligned}$$

- ▶  $x_j$  are observed per usual,  $y_i$  are individual demographics/features, and  $\varepsilon_{ij}$  is IID Type I EV. The key is that  $\nu_i$  is unobserved and mean zero (often normally distributed).
- ▶ Each individual draws a vector  $\beta_i / \mu_i$  (separately from  $\varepsilon_i$ ). Conditional on  $\beta_i / \mu_i$  each person follows an IIA logit model.
- ▶ However we integrate (or mix) over many such individuals giving us a **mixed logit** or **heirarchical model** (if you are a statistician)
- ▶ It helps to think about fixing  $\beta_i / \mu_i$  first and then integrating out over  $\varepsilon_i$

$$\sigma_{ij}(\theta) = \int \frac{\exp[x_j \beta_i]}{1 + \sum_k \exp[x_k \beta_i]} f(\beta_i | \theta) \approx \sum_{s=1}^S w_i^s \frac{\exp[x_j \beta_i^s]}{1 + \sum_k \exp[x_k \beta_i^s]}$$

- We can only **approximate** the integral with **weights**  $w_i^s$  and **nodes**  $\beta_i^s$
- We can allow for there to be two types of  $\beta_i$  in the population (high-type, low-type). **latent class model**.
  - We can allow  $\beta_i$  to follow an independent normal distribution for each component of  $x_{ij}$  such as  $\beta_i = \bar{\beta} + \Sigma \nu_i$ .
  - We can allow for correlated normal draws  $\beta_i \sim N(\mu_\beta, \Sigma_\beta)$ .
  - Can allow for non-normal distributions too (lognormal, exponential).
  - Why is normal so easy?

- ▶ The structure is extremely flexible but at a cost.
- ▶ We generally must perform the integration numerically.
- ▶ High-dimensional numerical integration is difficult. In fact, integration in dimension 8 or higher makes me very nervous.
- ▶ We need to be parsimonious in how many variables have unobservable heterogeneity.
- ▶ Again observed heterogeneity does not make life difficult so the more of that the better!
  - If we see individual income, education, distance to hospital, etc. we can always interact that with observed characteristics without doing any additional integration.

How does it work?

- ▶ Well we are mixing over individuals who conditional on  $\beta_i$  or  $\mu_i$  follow logit substitution patterns, however they may differ wildly in their  $\sigma_{ij}$  and hence their substitution patterns.
- ▶ For example if we are buying cameras: I may care a lot about price, you may care a lot about megapixels, and someone else may care mostly about zoom.
- ▶ The basic idea is that we need to explain the heteroskedasticity of  $Cov(\epsilon_i, \epsilon_\ell)$  what random coefficients do is let us use a basis from our  $X$ 's.
- ▶ If our  $X$ 's are able to span the space effectively, then an RC logit model can approximate any arbitrary RUM (such as probit) (McFadden and Train 2002).
- ▶ Of course if you have 1000 products and two random coefficients, you are asking for a lot.

At the level of an individual substitution patterns follow a plain logit:

$$\epsilon_{\sigma_j, p_j} = \frac{\partial \sigma_j}{\partial p_j} \cdot \frac{p_j}{\sigma_j} = \frac{p_j}{\sigma_j} \cdot \int \frac{\partial \sigma_{ij}}{\partial p_j} dF_i = \frac{p_j}{\sigma_j} \int \beta_i \cdot \sigma_{ij} \cdot (1 - \sigma_{ij}) dF_i$$

$$\epsilon_{\sigma_j, p_k} = \frac{\partial \sigma_j}{\partial p_k} \cdot \frac{p_k}{\sigma_j} = \frac{p_k}{\sigma_j} \cdot \int \frac{\partial \sigma_{ij}}{\partial p_k} dF_i = -\frac{p_k}{\sigma_j} \int \beta_i \cdot \sigma_{ij} \cdot \sigma_{ik} dF_i$$

Notation:  $\sigma_{ij}(\boldsymbol{\mu}_i) \equiv \sigma_{ij}$  is the individual choice probability and  $\sigma_j = \int \sigma_{ij} dF_i$  is its aggregate (market-level) counterpart.

At the level of an individual substitution patterns follow a plain logit:

$$D_{jk,i}(x) = \frac{\frac{\partial \sigma_{ik}}{\partial p_j}}{\left| \frac{\partial \sigma_{ij}}{\partial p_j} \right|} = \frac{\sigma_{ik}(x)}{1 - \sigma_{ij}(x)}$$

But at the aggregate level, we mix over heterogeneous individuals:

$$D_{jk}(x) = \int D_{jk,i}(x) w_i(z_j, z'_j, x) dF_i \quad \text{where} \quad w_i(z_j, z'_j, x) = \frac{q_{ij}(z'_j, x) - q_{ij}(z_j, x)}{q_j(z'_j, x) - q_j(z_j, x)}$$

Different interventions (price changes, product removals, quality changes, etc.) give different weighting schemes.

How do we estimate these models? (Derived from multinomial log-likelihood):

$$\theta_{MLE} = \arg \min_{\theta} - \sum_{i=1}^N \sum_{j=1}^J y_{ij} \ln \hat{\sigma}_{ij}(\theta)$$

- ▶ We need to perform numerical integration to get  $\hat{\sigma}_{ij}(\theta)$  and its derivative:  $\widehat{\frac{\partial \sigma_{ij}}{\partial \theta}}$ .
- ▶ Consistency
  - Technically for fixed number of MC draws the MSLE estimator is inconsistent. Why?
  - Very accurate integration is even more important!

Can also do a simulated GMM estimator with moment conditions:

$$\mathbb{E} [(y_{ij} - \hat{\sigma}_{ij}(\theta)) | z_{ij}] = 0$$

- ▶ We need to perform numerical integration to get  $\hat{\sigma}_{ij}(\theta)$  and its derivative:  $\widehat{\frac{\partial \sigma_{ij}}{\partial \theta}}$ .
- ▶ The **optimal instruments** in the sense of Chamberlain (1987) or Amemiya (1977) are the derivative of moments with respect to parameters (**scores**):  $z_{ij} = \frac{\partial \log \sigma_{ij}}{\partial \theta}(\theta)$ . (Much more on this when we get to BLP instruments next week.)
- ▶ The true scores are infeasible (they depend on  $\theta_0$ ).
  - At true scores: same FOC as MLE.
  - Simulated scores have same consistency issue as MSLE.

As an alternative we could work with the score function directly:

$$\sum_{i=1}^n \frac{\partial \widehat{\log \sigma_{ij}}}{\partial \theta}(\theta) = \sum_{i=1}^N \frac{1}{\widehat{\sigma_{ij}}(\theta)} \cdot \frac{\partial \widehat{\sigma_{ij}}}{\partial \theta}(\theta) = 0$$

- ▶ We can get unbiased estimated of derivative (linearity)
- ▶ We can't get unbiased estimate of  $\frac{1}{\widehat{\sigma_{ij}}(\theta)}$ 
  - Train's book suggests Accept-Reject
  - Maybe importance sampling would work?

How bad is the simulation error?

- ▶ Depends how small your shares are.
- ▶ Since you care about  $\log \sigma_{jt}$  when shares are small, tiny errors can be enormous.
- ▶ Often it is pretty bad.

## Estimation Details: Recommendations

You should read these separate notes, but my recommendations are:

### 1. Numerical integration:

- Use Gauss-Hermite quadrature rules to integrate over normal densities.
- Quadrature is great in low dimensions – but scales badly in high dimensions: if we need  $N_a$  points to accurately approximate the integral in  $d = 1$  then we need  $N_a^d$  points in dimension  $d$  (using the tensor product of quadrature rules).
- Follow Heiss and Winschel (JoE) and do **sparse grids** in medium dimensions
- Try <http://sparse-grids.de>.

### 2. Nonlinear optimization: with mixtures the problem is non-convex (ie: very hard)

- You **must** provide analytic gradients.
- You should use a quasi-Newton solver and may need to try several.
- Try multiple starting values.

### 3. Generalized Method of Moments

- You can read more about optimal IV in these settings.